

1. Write a java program to print Hello World.**CODE:**

```
class Hello{  
    public static void main(String args[]){  
        System.out.println("Hello, World!");  
    }  
}
```

OUTPUT:

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Hello.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Hello

Hello, World!

2. Write a java program to calculate area of circle.**CODE:**

```
public class Circle{  
    public static void main(String args[]){  
        double r=9.54;  
        double pi=3.14;  
        double a=pi*r*r;  
        System.out.println("Area of circle is "+a);  
    }  
}
```

OUTPUT:

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Circle.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Circle

Area of circle is 285.77642399999996

3. Write a java program to perform arithmetic operations on given number.**CODE:**

```
import java.io.DataInputStream;  
  
public class Ari{  
    public static void main(String args[]) throws Exception{  
        int a,b,add,sub,multiply,div,mod;  
        double res;  
  
        DataInputStream d1 = new DataInputStream(System.in);  
  
        System.out.print("Enter value of a:");
```

```
a=Integer.parseInt(d1.readLine());

System.out.print("Enter value of b:");

b=Integer.parseInt(d1.readLine());

add=a+b;

sub=a-b;

div=a/b;

multiply=a*b;

mod=a%b;

res=Math.pow(a,b);

System.out.println("Addition:"+add);

System.out.println("Subtraction:"+sub);

System.out.println("Multiplication:"+multiply);

System.out.println("Division:"+div);

System.out.println("Modulus:"+mod);

System.out.println("Power:"+res);

}

}
```

OUTPUT:

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Ari.java

Note: Ari.java uses or overrides a deprecated API.

Note: Recompile with -Xlint:deprecation for details.

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Ari

Enter value of a:4

Enter value of b:6

Addition:10

Subtraction:-2

Multiplication:24

Division:0

Modulus:4

Power:4096.0

4. Accept integer values for a, b and c which are coefficients of quadratic equation. Find the solution of quadratic equation. Consider Quadratic equation in this question is $2a^2+3b+c$.

CODE:

```
import java.util.Scanner;

public class Quadratic{

    public static void main(String args[]) throws Exception{

        Scanner d = new Scanner(System.in);

        System.out.println("To find solution of eq  $2a^2+3b+c$ ...");

        System.out.print("Enter the value for a:");

        int a = Integer.parseInt(d.nextLine());

        System.out.print("Enter the value for b:");

        int b = Integer.parseInt(d.nextLine());

        System.out.print("Enter the value for c:");

        int c = Integer.parseInt(d.nextLine());

        int r =  $2*a*a+3*b+c$ ;

        System.out.print("The solution of  $2a^2+3b+c$  is "+r);

    }

}
```

OUTPUT:

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Quadratic.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Quadratic

To find solution of eq $2a^2+3b+c$...

Enter the value for a:2

Enter the value for b:5

Enter the value for c:8

The solution of $2a^2+3b+c$ is 31

5. Write a java program to find maximum of two numbers. (if)

CODE:

```
import java.util.Scanner;

public class Find {

    public static void main(String args[]){

        Scanner s = new Scanner(System.in);

        System.out.print("Enter 1st integer number:");

        int a = s.nextInt();
```

```
System.out.print("Enter 2nd integer number:");

int b = s.nextInt();

if (a>b){

    System.out.println("Maximum number is "+a);

}

else{

    System.out.println("Maximum number is "+b);

}

}

}
```

OUTPUT:

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Find.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Find

Enter 1st integer number:5

Enter 2nd integer number:90

Maximum number is 90

6. Write a java program to find if number is even or odd. (if)**CODE:**

```
import java.util.Scanner;

public class OddEven {

    public static void main(String args[]){

        int a;

        Scanner s = new Scanner(System.in);

        System.out.print("Enter integer:");

        a = s.nextInt();

        if(a%2==0){

            System.out.println("Number is even.");

        }

        else{

            System.out.println("Number is odd.");

        }

    }

}
```

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac OddEven.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java OddEven

Enter integer:10

Number is even.

7. Write a java program to grade the students according to their grades.

CODE:

```
import java.util.Scanner;

public class Grade {

    public static void main(String args[]){

        Scanner s = new Scanner(System.in);

        int a;

        System.out.print("Enter total marks of student out of 300:");

        a = s.nextInt();

        if (a >= 270 && a <= 300){

            System.out.println("S grade.");

        }

        else if (a >= 250 && a < 270){

            System.out.println("A grade.");

        }

        else if (a >= 230 && a < 250){

            System.out.println("B grade.");

        }

        else if (a >= 180 && a < 230){

            System.out.println("C grade.");

        }

        else if (a >= 0 && a < 180){

            System.out.println("F grade.");

        }

        else {

            System.out.println("Invalid marks entered.");

        }

    }

}
```

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Grade.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Grade

Enter total marks of student out of 300:250

A grade.

8. Write a java program to find the factorial of given number using for loop.

CODE:

```
import java.util.Scanner;

public class Factorial {

    public static void main(String args[]){

        Scanner s = new Scanner(System.in);

        System.out.print("Enter the number to find the factorial:");

        int n = s.nextInt();

        if(n<0){

            System.out.println("Invalid Number!!");

        }

        else{

            long t=1;

            for(int i=1;i<=n;i++){

                t=i*t;

            }

            System.out.println("Factorial is "+t);

        }

    }

}
```

OUTPUT:

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Factorial.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Factorial

Enter the number to find the factorial:5

Factorial is 120

9. Accept two $n \times m$ matrices. Write a Java program to find addition of these matrices.

CODE:

```
import java.util.Scanner;

public class Matrice {

    public static void main(String args[]){

        Scanner s = new Scanner(System.in);

        System.out.print("Enter the value of m:");

        int m = s.nextInt();

        System.out.print("Enter the value of n:");

        int n = s.nextInt();

        int[][] arr1 = new int[m][n];

        int[][] arr2 = new int[m][n];

        int[][] sum = new int[m][n];

        System.out.println("Enter element of 1st matrix:");

        for(int i=0;i<m;i++){

            for(int j=0;j<n;j++){

                arr1[i][j] = s.nextInt();

            }

        }

        System.out.println("Enter element of 2nd matrix:");

        for(int i=0;i<m;i++){

            for(int j=0;j<n;j++){

                arr2[i][j] = s.nextInt();

            }

        }

        for(int i=0;i<m;i++){

            for(int j=0;j<n;j++){

                sum[i][j] = arr1[i][j]+arr2[i][j];

            }

        }

        System.out.println("Addition of two matrix is...");

        for(int i=0;i<m;i++){

            for(int j=0;j<n;j++){

                System.out.print(sum[i][j]+"\\t");
```

```
    }  
    System.out.println();  
}  
}  
}
```

OUTPUT:

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Matrice.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Matrice

Enter the value of m:3

Enter the value of n:3

Enter element of 1st matrix:

1 2 3 4 5 6 7 8 9

Enter element of 2nd matrix:

11 12 13 14 15 16 17 18 19

Addition of two matrix is...

12 14 16

18 20 22

24 26 28

10. Write a java program to calculate sum of all the odd numbers in the given range using while loop.

CODE:

```
import java.util.Scanner;  
  
public class Adding {  
    public static void main(String args[]){  
        Scanner s = new Scanner(System.in);  
        int num;  
        int n=0;  
        System.out.print("Enter the number:");  
        num = s.nextInt();  
        for(int i=0;i<num;i++){  
            if(i%2 != 0){  
                n = n+i;  
            }  
        }  
        System.out.print("Addition of all odd number:"+ n +" in a given range i.e "+num);
```



```

    }
}

```

OUTPUT:

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Adding.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Adding

Enter the number:10

Addition of all odd number:25 in a given range i.e 10

11. Write a java program to accept input from the user using command line arguments.**CODE:**

```

public class CmdLine{
    public static void main(String args[]) throws Exception {
        String name = args[0];
        int age = Integer.parseInt(args[1]);
        System.out.println("Hello, " + name + "! Your age is "+age+"");
    }
}

```

OUTPUT:

C:\Users\Rohit>javac CommandLine.java

C:\Users\Rohit>java CmdLine Atomic 18

Hello, Atomic! Your age is 18.

12. Write a java program using class to solve the following series. $Y=1+(2/3!)+(3/4!)+\dots$ **CODE:**

```

import java.util.Scanner;
public class Algorithm{
    static long factorial(int num) {
        long fact = 1;
        for (int i = 2; i <= num; i++) {
            fact *= i;}
        return fact;}
    static double calculateSum(int n) {
        double sum = 1.0;
        for (int i = 2; i <= n; i++) {
            sum += (double) i / factorial(i + 1);}
        return sum;}
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the number of terms: ");
        int n = scanner.nextInt();

        double result = calculateSum(n);
        System.out.println("Sum of the series is: " + result);
    }
}

```

OUTPUT:

Enter the number of terms: 5

Sum of the series is: 1.4986111111111111

13. Write a java program for printing the following output.

1

SYCS55

JAVA

PRACTICAL-1

22

333

4444

55555

666666

7777777

88888888

999999999

CODE:

```
public class NumPattern {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 9; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(i);  
            }  
            System.out.println();  
        }  
    }  
}
```

OUTPUT:

1

22

333

4444

55555

666666

7777777

88888888

999999999